SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name : AFX [HP]2

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Substance/Mixture : Catalyst for oil refining industry

Recommended restrictions on use : None known.

1.3 Details of the supplier of the safety data sheet

Company : Albemarle Catalysts Company BV
Site Amsterdam, Nieuwendammerkade 1-3
PO Box 37650 , Amsterdam
Netherlands

Telephone : +31.20.634.7000

Telefax : +31.20.634.7651

Contact person product safety : DEPARTMENT OF PRODUCT SAFETY

E-mail address : PRODUCTSAFETY@ALBEMARLE.COM

1.4 Emergency telephone number

Emergency telephone number : +32 (0) 70-233-201 (EUROPE)
(+1)225-344-7147 (US and WORLDWIDE)
+65-6733-1661 (ASIA PACIFIC)
+86-532-8388-9090 (CHINA)
+61 2 8014 4558 or 18000 74234 (Australia)

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Not a hazardous substance or mixture.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

Hazard statements : Not a hazardous substance or mixture.

Precautionary statements : Handle in accordance with good industrial hygiene and safety practice.

2.3 Other hazards

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.
Mechanical irritation of the eyes is possible.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number	Classification	Concentration (% w/w)
Aluminium oxide	1344-28-1 215-691-6 01-2119529248-35-0042		>= 10 - < 60
Zeolites	1318-02-1 215-283-8 01-2119429034-49-0004		>= 10 - < 50
Kaolin	1332-58-7 310-194-1		>= 5 - < 40
silicon dioxide	7631-86-9 231-545-4 01-2119379499-16-0027		>= 1 - < 30
lanthanum oxide	1312-81-8 215-200-5 01-2119487300-44-0000		>= 1 - < 10
aluminium orthophosphate	7784-30-7 232-056-9 01-2119971255-34-0001		>= 1 - < 5

For explanation of abbreviations see section 16.

SECTION 4: First aid measures

4.1 Description of first aid measures

General advice	:	First Aid responders should pay attention to self-protection and use the recommended protective clothing Ensure that eyewash stations and safety showers are close to the workstation location.
If inhaled	:	Remove person to fresh air and keep comfortable for breathing. If breathing is difficult, give oxygen. If breathing is irregular or stopped, administer artificial respiration. Keep the victim calm and in a semi-upright position. Call a physician or poison control centre immediately.
In case of skin contact	:	Wash with plenty of soap and water. If symptoms persist, call a physician.
In case of eye contact	:	Rinse immediately with plenty of water, also under the eyelids. Seek medical advice.
If swallowed	:	Clean mouth with water and drink afterwards plenty of water. Obtain medical attention.

4.2 Most important symptoms and effects, both acute and delayed

Symptoms	:	Mechanical irritation of the eyes is possible.
Risks	:	See Section 2.

4.3 Indication of any immediate medical attention and special treatment needed

Treatment	:	Treat symptomatically. For specialist advice physicians should contact the Poisons Information Service.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media	:	Not combustible. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	:	No information available.

5.2 Special hazards arising from the substance or mixture

Specific hazards during fire-	:	In the event of fire and/or explosion do not breathe fumes.

fighting

5.3 Advice for firefighters

Special protective equipment for firefighters	:	Wear full protective clothing and self-contained breathing apparatus.
Further information	:	Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions	:	Avoid dust formation. Avoid breathing dust. Ensure adequate ventilation. Keep people away from and upwind of spill/leak.

6.2 Environmental precautions

Environmental precautions	:	Avoid release to the environment. Do not allow contact with soil, surface or ground water. Prevent further leakage or spillage if safe to do so.

6.3 Methods and material for containment and cleaning up

Methods for cleaning up	:	Pick up and transfer to properly labelled containers. Clean contaminated floors and objects thoroughly while observing environmental regulations. Residue may be washed down with water.

6.4 Reference to other sections

For personal protection see section 8., For disposal considerations see section 13.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advice on safe handling	:	Avoid creating dust. Do not breathe dust. Avoid contact with skin and eyes. Provide sufficient air exchange and/or exhaust in work rooms. Wear personal protective equipment. For personal protection see section 8. In general, emissions are controlled and prevented by implementing an appropriate management system, including regular informing and training workers.
Hygiene measures	:	Handle in accordance with good industrial hygiene and safety

practice.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : Keep container tightly closed in a dry and well-ventilated place.

7.3 Specific end use(s)

Specific use(s) : No data available

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Derived No Effect Level (DNEL) according to Regulation (EC) No. 1907/2006:

Substance name	End Use	Exposure routes	Potential health effects	Value
Zeolites	Consumers	Inhalation	Long-term local effects	0.003 mg/m3
	Consumers	Dermal	Long-term systemic effects	1.25 mg/kg bw/day
	Workers	Dermal	Long-term systemic effects	2.5 mg/kg bw/day
	Consumers	Oral	Long-term systemic effects	1.25 mg/kg bw/day
	Workers	Inhalation	Long-term local effects	3 mg/m3
Aluminium oxide	Consumers	Oral	Long-term systemic effects	6.58 mg/kg bw/day
	Workers	Inhalation	Long-term local effects	15.63 mg/m3
	Workers	Inhalation	Long-term systemic effects	15.63 mg/m3
silicon dioxide	Workers	Inhalation	Long-term systemic effects	4 mg/m3
aluminium orthophosphate	Consumers	Oral	Long-term systemic effects	0.71 mg/kg bw/day
	Consumers	Inhalation	Long-term systemic effects	2.67 mg/m3
	Workers	Inhalation	Long-term systemic effects	4.98 mg/m3

8.2 Exposure controls

Engineering measures

Provide sufficient air exchange and/or exhaust in work rooms.
Mechanical ventilation is recommended.
Local exhaust is needed at source of dust.
Training in proper use of PPE provided and ensured via an inspection policy.

Personal protective equipment

Eye protection	:	Wear safety glasses with side shields or goggles.
Hand protection Material	:	Wear suitable gloves resistant to chemical penetration.
Skin and body protection	:	Wear suitable protective clothing. Long sleeved clothing Full protective suit
Respiratory protection	:	In case of inadequate ventilation wear respiratory protection. In the case of dust or aerosol formation use respirator with an approved filter.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Appearance	:	solid
Colour	:	No data available
Odour	:	odourless
Odour Threshold	:	Negligible
pH	:	No data available
Melting point/freezing point	:	No data available
Boiling point/boiling range	:	study scientifically unjustified Melting point > 300 °C
Flash point	:	Not applicable, solid
Evaporation rate	:	Negligible
Flammability (solid, gas)	:	The product is not flammable.
Burning number	:	No data available
Upper explosion limit / Upper flammability limit	:	Not applicable
Lower explosion limit / Lower flammability limit	:	Not applicable
Vapour pressure	:	Negligible
Relative vapour density	:	Not applicable Melting point > 300 °C

Relative density	:	No data available
Density	:	No data available
Bulk density	:	600 - 1,200 kg/m3(compacted, bulk)
Solubility(ies)		
Water solubility	:	Negligible
Solubility in other solvents	:	No data available
Partition coefficient: n-octanol/water	:	Not applicable inorganic
Auto-ignition temperature	:	No data available
Decomposition temperature	:	Stable up to the melting point.
Viscosity		
Viscosity, dynamic	:	Not applicable solid
Viscosity, kinematic	:	Not applicable solid
Explosive properties	:	No chemical groups associated to explosive properties.
Oxidizing properties	:	No chemical groups associated to oxidising properties.

9.2 Other information

Flammability (liquids)	:	Not applicable
Self-heating substances	:	The substance or mixture is not classified as self heating.
Surface tension	:	Not applicable, solid
Sublimation point	:	Not applicable
Molecular weight	:	No data available

SECTION 10: Stability and reactivity

10.1 Reactivity

No dangerous reaction known under conditions of normal use.

10.2 Chemical stability

Stable up to the melting point.

10.3 Possibility of hazardous reactions

Hazardous reactions : No dangerous reaction known under conditions of normal use.

10.4 Conditions to avoid

Conditions to avoid : Avoid dust formation.
Protect from moisture.
Avoid extremely high temperatures.

10.5 Incompatible materials

Materials to avoid : None.

10.6 Hazardous decomposition products

No decomposition if used as directed.

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Acute toxicity

Components:

Aluminium oxide:

Acute oral toxicity : LD50 (Rat, male and female): > 5,000 mg/kg
Remarks: No mortality observed at this dose.

Acute dermal toxicity : Remarks: study scientifically unjustified

Zeolites:

Acute oral toxicity : LD50 (Rat, male and female): > 5,000 mg/kg
Method: OECD Test Guideline 401
Remarks: No mortality observed at this dose.

Acute inhalation toxicity : LC50 (Rat, male and female): > 3.35 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Remarks: An LC50/inhalation/4h/rat could not be determined because no mortality of rats was observed at the maximum achievable concentration.

Acute dermal toxicity : LD50 (Rabbit, female): > 2,000 mg/kg
Remarks: No mortality observed at this dose.

Kaolin:

Acute oral toxicity : Remarks: No data available

Acute inhalation toxicity : Remarks: No data available

Acute dermal toxicity : Remarks: No data available

silicon dioxide:

Acute oral toxicity : LD50 (Rat, male and female): > 5,000 mg/kg
Method: OECD Test Guideline 401
Remarks: No mortality observed at this dose.

Acute inhalation toxicity : LC50 (Rat, male and female): > 0.14 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Remarks: An LC50/inhalation/4h/rat could not be determined because no mortality of rats was observed at the maximum achievable concentration.

Acute dermal toxicity : LD50 (Rabbit, male): > 5,000 mg/kg
Remarks: No mortality observed at this dose.

lanthanum oxide:

Acute oral toxicity : LD50 (Rat): > 2,000 mg/kg
Remarks: Expert judgement

Acute inhalation toxicity : LC50 (Rat, male and female): > 5.3 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Remarks: An LC50/inhalation/4h/rat could not be determined because no mortality of rats was observed at the maximum achievable concentration.

Acute dermal toxicity : LD50 (Rat, male and female): > 2,000 mg/kg
Method: According to a standard method
Remarks: No mortality observed at this dose.

aluminium orthophosphate:

Acute oral toxicity : LD50 (Rat, female): > 2,000 mg/kg
Method: OECD Test Guideline 420
Remarks: No mortality observed at this dose.

Acute dermal toxicity : Remarks: study scientifically unjustified

Skin corrosion/irritation

<u>Components:</u>

Aluminium oxide:

Species : Rabbit
Result : No skin irritation

Zeolites:

Species : Rabbit
Method : OECD Test Guideline 404
Result : No skin irritation

Kaolin:

Remarks : No data available

silicon dioxide:

Species : Rabbit
Method : OECD Test Guideline 404
Result : No skin irritation

lanthanum oxide:

Species : Rabbit
Method : Draize Test
Result : No skin irritation

aluminium orthophosphate:

Species : reconstructed human epidermis (RhE)
Method : OECD Test Guideline 439
Result : negative

Species : reconstructed human epidermis (RhE)
Method : OECD Test Guideline 431
Result : negative

Serious eye damage/eye irritation

Product:

Remarks : Mechanical irritation of the eyes is possible.

Components:

Aluminium oxide:

Species : Rabbit
Method : Draize Test
Result : No eye irritation

Zeolites:

Species : Rabbit
Method : OECD Test Guideline 405
Result : No eye irritation

Kaolin:

Remarks : No data available

silicon dioxide:

Species : Rabbit
Method : OECD Test Guideline 405
Result : No eye irritation

lanthanum oxide:

Species : Rabbit
Method : Draize Test
Result : No eye irritation

aluminium orthophosphate:

Species : Rabbit
Method : OECD Test Guideline 405
Result : No eye irritation

Respiratory or skin sensitisation

Components:

Aluminium oxide:

Exposure routes : Skin contact
Species : Guinea pig
Result : Did not cause sensitisation on laboratory animals.

Exposure routes : intratracheal
Species : Mouse
Result : Did not cause sensitisation on laboratory animals.

Zeolites:

Test Type : Buehler Test
Exposure routes : Skin contact
Species : Guinea pig
Method : OECD Test Guideline 406
Result : Did not cause sensitisation on laboratory animals.

Kaolin:

Remarks : No data available

silicon dioxide:

Remarks : study scientifically unjustified

lanthanum oxide:

Test Type : Maximisation Test
Exposure routes : Skin contact
Species : Guinea pig
Method : OECD Test Guideline 406
Result : Did not cause sensitisation on laboratory animals.

aluminium orthophosphate:

Test Type : Local lymph node assay (LLNA)
Exposure routes : Skin contact
Species : Mouse
Method : OECD Test Guideline 429
Result : Did not cause sensitisation on laboratory animals.

Germ cell mutagenicity

Components:

Aluminium oxide:

Genotoxicity in vitro : Result: Positive results were obtained in some in vitro tests.
Remarks: Expert judgement and weight of evidence determination.

Genotoxicity in vivo : Test Type: Chromosome aberration test in vivo
Species: Rat (female)
Application Route: Oral
Method: OECD Test Guideline 475
Result: negative

Test Type: In vivo micronucleus test
Species: Rat (male and female)
Application Route: Oral
Method: OECD Test Guideline 474
Result: negative
Test substance: Similar substance

Germ cell mutagenicity- Assessment : Not classified due to data which are conclusive although insufficient for classification.

Zeolites:

Genotoxicity in vitro : Test Type: Microbial mutagenesis assay (Ames test)
Test system: Salmonella typhimurium; Escherischia coli
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative

Test Type: Chromosome aberration test in vitro
Test system: Chinese hamster ovary cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 473
Result: positive

Test Type: In vitro mammalian cell gene mutation test
Test system: mouse lymphoma cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476

		Result: negative
Genotoxicity in vivo	:	Test Type: Chromosome aberration test in vivo Species: Rat (male) Application Route: Oral Result: negative
		Test Type: dominant lethal test Species: Rat (male) Application Route: Oral Result: negative
		Test Type: In vivo micronucleus test Species: Mouse (male and female) Application Route: Oral Method: OECD Test Guideline 474 Result: negative
Germ cell mutagenicity- Assessment	:	Not classified due to data which are conclusive although insufficient for classification.

Kaolin:

Germ cell mutagenicity- Assessment	:	Not classified due to lack of data.

silicon dioxide:

Genotoxicity in vitro	:	Test Type: Microbial mutagenesis assay (Ames test) Test system: Salmonella typhimurium Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 471 Result: negative
		Test Type: Chromosome aberration test in vitro Test system: Chinese hamster ovary cells Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 473 Result: negative
		Test Type: In vitro mammalian cell gene mutation test Test system: Chinese hamster ovary cells Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 476 Result: negative
Genotoxicity in vivo	:	Test Type: gene mutation test Species: Rat (male) Application Route: inhalation (dust/mist/fume) Result: negative
Germ cell mutagenicity- As-	:	Not classified due to data which are conclusive although insuf-

sessment		ficient for classification.

lanthanum oxide:

Genotoxicity in vitro	:	Test Type: Microbial mutagenesis assay (Ames test) Test system: Salmonella typhimurium Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 471 Result: negative
		Test Type: Chromosome aberration test in vitro Test system: Chinese hamster ovary cells Metabolic activation: with and without metabolic activation Result: negative
		Test Type: In vitro mammalian cell gene mutation test Test system: Chinese hamster ovary cells Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 476 Result: negative
Genotoxicity in vivo	:	Test Type: unscheduled DNA synthesis assay Species: Rat (male) Application Route: Intravenous Method: OECD Test Guideline 486 Result: negative
		Test Type: In vivo micronucleus test Species: Rat (male) Application Route: Intravenous Method: OECD Test Guideline 474 Result: negative
Germ cell mutagenicity- Assessment	:	Not classified due to data which are conclusive although insufficient for classification.

aluminium orthophosphate:

Genotoxicity in vitro	:	Test Type: Microbial mutagenesis assay (Ames test) Test system: Salmonella typhimurium; Escherischia coli Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 471 Result: negative
		Test Type: Chromosome aberration test in vitro Test system: Human lymphocytes Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 473 Result: negative
		Test Type: In vitro mammalian cell gene mutation test Test system: mouse lymphoma cells

		Metabolic activation: with metabolic activation Method: OECD Test Guideline 476 Result: negative
Genotoxicity in vivo	:	Remarks: study scientifically unjustified
Germ cell mutagenicity- Assessment	:	Not classified due to data which are conclusive although insufficient for classification.

Carcinogenicity

Components:

Aluminium oxide:

Remarks	:	Weight of evidence does not support classification as a carcinogen
Carcinogenicity - Assessment	:	Not classified due to data which are conclusive although insufficient for classification.

Zeolites:

Species	:	Rat, male and female
Application Route	:	oral (feed)
Exposure time	:	2 Years
NOAEL	:	>= 60 mg/kg bw/day
Result	:	Animal testing did not show any carcinogenic effects.
Carcinogenicity - Assessment	:	Not classified due to data which are conclusive although insufficient for classification.

Kaolin:

Carcinogenicity - Assessment	:	Not classified due to lack of data.

silicon dioxide:

Species	:	Rat, male and female
Application Route	:	Intrapleural
Exposure time	:	2 Years
Result	:	Animal testing did not show any carcinogenic effects.
Species	:	Rat, male and female
Application Route	:	oral (feed)
Exposure time	:	2 Years
NOAEL	:	>= 1,800 mg/kg bw/day
Result	:	Animal testing did not show any carcinogenic effects.
Carcinogenicity - Assessment	:	Not classified due to data which are conclusive although insufficient for classification.

lanthanum oxide:

Carcinogenicity - Assessment : Not classified due to lack of data.

aluminium orthophosphate:

Remarks : Weight of evidence does not support classification as a carcinogen

Carcinogenicity - Assessment : Not classified due to data which are conclusive although insufficient for classification.

Reproductive toxicity

<u>Components:</u>

Aluminium oxide:

Effects on fertility : Species: Rat, male and female
Application Route: oral (gavage)
General Toxicity - Parent: NOAEL: 200 mg/kg bw/day
General Toxicity F1: NOAEL: >= 1,000 mg/kg bw/day
Fertility: LOAEC: >= 1,000 mg/kg bw/day
Method: OECD Test Guideline 422
Result: Animal testing did not show any effects on fertility.
Remarks: Information given is based on data obtained from similar substances.

Effects on foetal development : Species: Rat, female
Application Route: oral (drinking water)
General Toxicity Maternal: NOAEL: 300 mg/kg bw/day
Developmental Toxicity: NOAEL: 300 mg/kg bw/day
Result: No evidence of adverse effects on sexual function and fertility, or on development, based on animal experiments.
Remarks: Information given is based on data obtained from similar substances.

Reproductive toxicity - Assessment : Not classified due to data which are conclusive although insufficient for classification.

Zeolites:

Effects on fertility : Remarks: Weight of evidence does not support classification for reproductive toxicity

Effects on foetal development : Test Type: Pre-/postnatal development
Species: Rat, female
Application Route: oral (gavage)
General Toxicity Maternal: NOAEL: >= 1,600 mg/kg bw/day
Developmental Toxicity: NOAEL: >= 1,600 mg/kg bw/day
Remarks: No evidence of adverse effects on sexual function and fertility, or on development, based on animal experiments.

Reproductive toxicity - Assessment	:	Not classified due to data which are conclusive although insufficient for classification.

Kaolin:

Reproductive toxicity - Assessment	:	Not classified due to lack of data.

silicon dioxide:

Effects on fertility	:	Remarks: study scientifically unjustified
Effects on foetal development	:	Test Type: Pre-/postnatal development Species: Rat, female Application Route: oral (gavage) General Toxicity Maternal: NOAEL: >= 1,350 mg/kg bw/day Developmental Toxicity: NOAEL: >= 1,350 mg/kg bw/day Result: No evidence of adverse effects on sexual function and fertility, or on development, based on animal experiments.
Reproductive toxicity - Assessment	:	Not classified due to data which are conclusive although insufficient for classification.

lanthanum oxide:

Effects on fertility	:	Species: Mouse, female Application Route: oral (drinking water) General Toxicity - Parent: NOAEL: 40 mg/kg bw/day Fertility: LOAEC: 40 mg/kg bw/day Result: No indication of any effect on fertility and development beyond the dose levels of apparent maternal toxicity.
Effects on foetal development	:	Species: Mouse, female General Toxicity Maternal: NOAEL: 40 mg/kg bw/day Developmental Toxicity: NOAEL: 40 mg/kg bw/day Result: No evidence of adverse effects on sexual function and fertility, or on development, based on animal experiments.
Reproductive toxicity - Assessment	:	Not classified due to data which are conclusive although insufficient for classification.

aluminium orthophosphate:

Effects on fertility	:	Test Type: Two-generation study Species: Rat, male and female Application Route: oral (drinking water) General Toxicity - Parent: LOAEL: 600 ppm General Toxicity F1: LOAEL: 600 ppm Fertility: LOAEL: 600 ppm Method: OECD Test Guideline 416 Result: Animal testing did not show any effects on fertility. Remarks: Information given is based on data obtained from similar substances.

Effects on foetal development	:	Species: Rat, female General Toxicity Maternal: NOAEL: 300 mg/kg bw/day Developmental Toxicity: 300 mg/kg bw/day Result: No evidence of adverse effects on sexual function and fertility, or on development, based on animal experiments. Remarks: Information given is based on data obtained from similar substances.
Reproductive toxicity - Assessment	:	Not classified due to data which are conclusive although insufficient for classification.

STOT - single exposure

Components:

Aluminium oxide:

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

Zeolites:

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

Kaolin:

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

silicon dioxide:

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

lanthanum oxide:

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

aluminium orthophosphate:

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

STOT - repeated exposure

Components:

Aluminium oxide:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Zeolites:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Kaolin:

Assessment : Not classified due to lack of data.

silicon dioxide:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

lanthanum oxide:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

aluminium orthophosphate:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity

Components:

Aluminium oxide:

Species : Rat, male and female
LOAEC : 70 mg/m³
Application Route : inhalation (dust/mist/fume)
Exposure time : 90 d
Remarks : No significant adverse effects were reported

Zeolites:

Species : Rat, male and female
NOAEL : >= 500 mg/kg bw/day
Application Route : oral (feed)
Exposure time : 90 d
Remarks : No significant adverse effects were reported

Species : Monkey, male and female
LOAEC : 1 mg/m³
Application Route : inhalation (dust/mist/fume)
Exposure time : 2 years
Remarks : No significant adverse effects were reported

silicon dioxide:

Species : Rat, male and female

NOAEL	:	>= 4000 mg/kg bw/day
Application Route	:	oral (feed)
Exposure time	:	90 d
Remarks	:	No significant adverse effects were reported

lanthanum oxide:

Species	:	Rat, male and female
NOAEL	:	>= 974 mg/kg bw/day
Application Route	:	oral (feed)
Exposure time	:	90 d
Method	:	OECD Test Guideline 408
Test substance	:	Similar substance
Remarks	:	No significant adverse effects were reported

aluminium orthophosphate:

Remarks	:	Expert judgement and weight of evidence determination. No significant adverse effects were reported

Aspiration toxicity

Product:

Not classified due to data which are conclusive although insufficient for classification.

Further information

Product:

Remarks	:	This product contains quartz originating from the clay used for catalyst manufacturing. Quartz has been classified by IARC (International Agency for Research on Cancer) as carcinogenic to humans by inhalation (Group 1). Furthermore, quartz can cause silicosis and other lung diseases on prolonged exposure by inhalation. Catalysts manufacturers have therefore decided to indicate the presence of a carcinogenic substance in their product, in accordance with the relevant EU regulations. However, the amount of quartz is low, and the quartz is generally not available in the form of free inhalable particles. Measurements have shown that under normal working conditions operators are not exposed to appreciable amounts of respirable quartz particulates. Nevertheless, the user is responsible for controlling the working environment according to local regulations.

SECTION 12: Ecological information

12.1 Toxicity

Product:

No data is available on the product itself.

Components:

Aluminium oxide:

Toxicity to fish	:	Remarks: Based on transformation/dissolution testing and data from soluble metal compounds Aquatic toxicity is unlikely due to low solubility.
Toxicity to daphnia and other aquatic invertebrates	:	Remarks: Based on transformation/dissolution testing and data from soluble metal compounds Aquatic toxicity is unlikely due to low solubility.
Toxicity to algae/aquatic plants	:	Remarks: Based on transformation/dissolution testing and data from soluble metal compounds Aquatic toxicity is unlikely due to low solubility.
Toxicity to microorganisms	:	EC50 (activated sludge): > 1,000 mg/l End point: Respiration inhibition Exposure time: 3 h Test Type: static test Analytical monitoring: no Test substance: Similar substance Method: OECD Test Guideline 209
Toxicity to fish (Chronic toxicity)	:	Remarks: Based on transformation/dissolution testing and data from soluble metal compounds Aquatic toxicity is unlikely due to low solubility.
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	Remarks: Based on transformation/dissolution testing and data from soluble metal compounds Aquatic toxicity is unlikely due to low solubility.

Ecotoxicology Assessment

Acute aquatic toxicity	:	No toxicity at the limit of solubility
Chronic aquatic toxicity	:	No toxicity at the limit of solubility

Zeolites:

Toxicity to fish	:	LL50 (Pimephales promelas (fathead minnow)): > 680 mg/l Exposure time: 96 h Test Type: static test Analytical monitoring: yes Method: According to a standard method
Toxicity to daphnia and other aquatic invertebrates	:	EL50 (Daphnia magna (Water flea)): > 1,000 mg/l End point: Immobilization Exposure time: 24 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 202

		Remarks: No toxicity at the limit of solubility
Toxicity to algae/aquatic plants	:	EC50 (Desmodesmus subspicatus (green algae)): 130 mg/l End point: Growth rate Exposure time: 72 h Test Type: static test Analytical monitoring: no Method: OECD Test Guideline 201 EC10 (Desmodesmus subspicatus (green algae)): 18 mg/l End point: Growth rate Exposure time: 72 h Test Type: static test Analytical monitoring: no Method: OECD Test Guideline 201
Toxicity to microorganisms	:	EC50 (activated sludge): 950 mg/l End point: Respiration inhibition Exposure time: 16 h Test Type: static test Analytical monitoring: no Method: According to a standard method
Toxicity to fish (Chronic toxicity)	:	NOEC: > 86.7 mg/l End point: Hatching/Post-hatching survival Exposure time: 30 d Species: Pimephales promelas (fathead minnow) Test Type: flow-through test Analytical monitoring: yes Method: According to a standard method
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC: 32 mg/l End point: Reproduction Exposure time: 21 d Species: Daphnia magna (Water flea) Test Type: semi-static test Analytical monitoring: no Method: OECD Test Guideline 211

Ecotoxicology Assessment

Acute aquatic toxicity	:	Not classified due to data which are conclusive although insufficient for classification.
Chronic aquatic toxicity	:	Not classified due to data which are conclusive although insufficient for classification.

Kaolin:

Ecotoxicology Assessment

Acute aquatic toxicity	:	Not classified due to lack of data.

Chronic aquatic toxicity	:	Not classified due to lack of data.

silicon dioxide:

Toxicity to fish	:	LL50 (Danio rerio (zebra fish)): >= 10,000 mg/l Exposure time: 96 h Test Type: static test Analytical monitoring: no Method: OECD Test Guideline 203 Remarks: No toxicity at the limit of solubility
Toxicity to daphnia and other aquatic invertebrates	:	EL50 (Daphnia magna (Water flea)): >= 1,000 mg/l End point: Immobilization Exposure time: 48 h Test Type: static test Analytical monitoring: no Test substance: Similar substance Method: OECD Test Guideline 202 Remarks: No toxicity at the limit of solubility
Toxicity to algae/aquatic plants	:	EL50 (Desmodesmus subspicatus (green algae)): >= 10,000 mg/l End point: Growth rate Exposure time: 72 h Test Type: static test Analytical monitoring: no Test substance: Similar substance Method: OECD Test Guideline 201 Remarks: No toxicity at the limit of solubility NOELR (Desmodesmus subspicatus (green algae)): >= 10,000 mg/l End point: Growth rate Exposure time: 72 h Test Type: static test Analytical monitoring: no Test substance: Similar substance Method: OECD Test Guideline 201 Remarks: No toxicity at the limit of solubility
Toxicity to microorganisms	:	 Remarks: Aquatic toxicity is unlikely due to low solubility.
Toxicity to fish (Chronic toxicity)	:	NOEC: 86.03 mg/l Exposure time: 30 d Species: Fish Method: QSAR
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC: > 100 mg/l Exposure time: 21 d Species: Daphnia magna (Water flea) Method: QSAR

Ecotoxicology Assessment

Acute aquatic toxicity : No toxicity at the limit of solubility

Chronic aquatic toxicity : No toxicity at the limit of solubility

lanthanum oxide:

Toxicity to fish : LL50 (Danio rerio (zebra fish)): > 100 mg/l
Exposure time: 96 h
Test Type: static test
Analytical monitoring: no
Method: OECD Test Guideline 203
Remarks: No toxicity at the limit of solubility

Toxicity to daphnia and other aquatic invertebrates : EL50 (Daphnia magna (Water flea)): > 100 mg/l
End point: Immobilization
Exposure time: 48 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 202
Remarks: No toxicity at the limit of solubility

Toxicity to algae/aquatic plants : EL50 (Desmodesmus subspicatus (green algae)): > 100 mg/l
End point: Growth rate
Exposure time: 72 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 201
Remarks: No toxicity at the limit of solubility

EL10 (Desmodesmus subspicatus (green algae)): 91 mg/l
End point: Growth rate
Exposure time: 72 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 201
Remarks: No toxicity at the limit of solubility

Toxicity to microorganisms : EC50 (activated sludge): 390 mg/l
End point: Respiration inhibition
Exposure time: 3 h
Test Type: static test
Analytical monitoring: no
Test substance: Similar substance
Method: OECD Test Guideline 209

Toxicity to fish (Chronic toxicity) : NOELR: > 100 mg/l
End point: mortality
Exposure time: 21 d
Species: Cyprinus carpio (Carp)

		Test Type: semi-static test Analytical monitoring: yes Test substance: Similar substance Method: OECD Test Guideline 204 Remarks: No toxicity at the limit of solubility
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOELR: > 100 mg/l End point: Reproduction Exposure time: 21 d Species: Daphnia magna (Water flea) Test Type: semi-static test Analytical monitoring: yes Method: OECD Test Guideline 211 Remarks: No toxicity at the limit of solubility

Ecotoxicology Assessment

Acute aquatic toxicity	:	No toxicity at the limit of solubility
Chronic aquatic toxicity	:	No toxicity at the limit of solubility

aluminium orthophosphate:

Toxicity to fish	:	Remarks: Based on transformation/dissolution testing and data from soluble metal compounds Aquatic toxicity is unlikely due to low solubility.
Toxicity to daphnia and other aquatic invertebrates	:	Remarks: Based on transformation/dissolution testing and data from soluble metal compounds Aquatic toxicity is unlikely due to low solubility.
Toxicity to algae/aquatic plants	:	Remarks: Based on transformation/dissolution testing and data from soluble metal compounds Aquatic toxicity is unlikely due to low solubility.
Toxicity to microorganisms	:	EC50 (activated sludge): > 1,000 mg/l End point: Respiration inhibition Exposure time: 3 h Test Type: static test Analytical monitoring: no Test substance: Similar substance Method: OECD Test Guideline 209
Toxicity to fish (Chronic toxicity)	:	Remarks: Based on transformation/dissolution testing and data from soluble metal compounds Aquatic toxicity is unlikely due to low solubility.
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	Remarks: Based on transformation/dissolution testing and data from soluble metal compounds Aquatic toxicity is unlikely due to low solubility.

Ecotoxicology Assessment

Acute aquatic toxicity : Not classified due to data which are conclusive although insufficient for classification.

No toxicity at the limit of solubility

Chronic aquatic toxicity : Not classified due to data which are conclusive although insufficient for classification.

No toxicity at the limit of solubility

12.2 Persistence and degradability

Components:

Aluminium oxide:

Biodegradability : Remarks: The methods for determining biodegradability are not applicable to inorganic substances.

Stability in water : Remarks: Not susceptible to hydrolysis.

Photodegradation : Remarks: Not susceptible to photodegradation.

Zeolites:

Biodegradability : Remarks: The methods for determining biodegradability are not applicable to inorganic substances.

Stability in water : Remarks: Not susceptible to hydrolysis.

Photodegradation : Remarks: Not susceptible to photodegradation.

Kaolin:

Biodegradability : Remarks: The methods for determining biodegradability are not applicable to inorganic substances.

Stability in water : Remarks: Not susceptible to hydrolysis.

Photodegradation : Remarks: Not susceptible to photodegradation.

silicon dioxide:

Biodegradability : Remarks: The methods for determining biodegradability are not applicable to inorganic substances.

Stability in water : Remarks: Not susceptible to hydrolysis.

Photodegradation : Remarks: Not susceptible to photodegradation.

lanthanum oxide:

Biodegradability	:	Remarks: The methods for determining biodegradability are not applicable to inorganic substances.
Stability in water	:	Remarks: Not susceptible to hydrolysis.
Photodegradation	:	Remarks: Not susceptible to photodegradation.

aluminium orthophosphate:

Biodegradability	:	Remarks: The methods for determining biodegradability are not applicable to inorganic substances.
Stability in water	:	Remarks: Not susceptible to hydrolysis.
Photodegradation	:	Remarks: Not susceptible to photodegradation.

12.3 Bioaccumulative potential

Components:

Aluminium oxide:

Bioaccumulation	:	Remarks: Bioaccumulation is unlikely.
Partition coefficient: n-octanol/water	:	Remarks: study scientifically unjustified inorganic

Zeolites:

Bioaccumulation	:	Remarks: Bioaccumulation is unlikely.
Partition coefficient: n-octanol/water	:	Remarks: study scientifically unjustified inorganic

Kaolin:

Bioaccumulation	:	Remarks: Bioaccumulation is unlikely.
Partition coefficient: n-octanol/water	:	Remarks: study scientifically unjustified inorganic

silicon dioxide:

Bioaccumulation	:	Remarks: Bioaccumulation is unlikely.
Partition coefficient: n-octanol/water	:	Remarks: study scientifically unjustified inorganic

lanthanum oxide:

Bioaccumulation	:	Remarks: Bioaccumulation is unlikely.
Partition coefficient: n-octanol/water	:	Remarks: study scientifically unjustified inorganic

aluminium orthophosphate:

Bioaccumulation : Remarks: Bioaccumulation is unlikely.

Partition coefficient: n-octanol/water : Remarks: study scientifically unjustified inorganic

12.4 Mobility in soil

Components:

Aluminium oxide:

Distribution among environmental compartments : Remarks: Is not likely mobile in the environment due to its low water solubility and propensity to bind soil particles.

Zeolites:

Distribution among environmental compartments : Remarks: Expected to be essentially immobile in soils.

silicon dioxide:

Distribution among environmental compartments : log Koc: 1.34
Method: QSAR
Remarks: Will be likely mobile in the environment.

lanthanum oxide:

Distribution among environmental compartments : Medium: Soil
log Koc: 6.36
Method: OECD Test Guideline 106
Remarks: Immobile in soils.

aluminium orthophosphate:

Distribution among environmental compartments : Remarks: Is not likely mobile in the environment due to its low water solubility and propensity to bind soil particles.

12.5 Results of PBT and vPvB assessment

Components:

Aluminium oxide:

Assessment : PBT/vPvB assessment is not applicable to inorganic substances.

Zeolites:

Assessment : PBT/vPvB assessment is not applicable to inorganic substances.

Kaolin:

Assessment : PBT/vPvB assessment is not applicable to inorganic substances.

silicon dioxide:

Assessment : PBT/vPvB assessment is not applicable to inorganic substances.

lanthanum oxide:

Assessment : PBT/vPvB assessment is not applicable to inorganic substances.

aluminium orthophosphate:

Assessment : PBT/vPvB assessment is not applicable to inorganic substances.

12.6 Other adverse effects

No data available

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product : Dispose of contents/ container to an approved facility in accordance with local, regional, national and international regulations.

Contaminated packaging : Refer to manufacturer/ supplier for information on recovery/ recycling.

SECTION 14: Transport information

14.1 UN number

Not regulated as a dangerous good

14.2 UN proper shipping name

Not regulated as a dangerous good

14.3 Transport hazard class(es)

Not regulated as a dangerous good

14.4 Packing group

Not regulated as a dangerous good

14.5 Environmental hazards

Not regulated as a dangerous good

14.6 Special precautions for user

Not applicable

14.7 Transport in bulk according to Annex II of Marpol and the IBC Code

Not applicable for product as supplied.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

The components of this product are reported in the following inventories:

TCSI	:	On the inventory, or in compliance with the inventory
TSCA	:	On or in compliance with the active portion of the TSCA inventory
AICS	:	On the inventory, or in compliance with the inventory
DSL	:	This product contains the following components that are not on the Canadian DSL nor NDSL. Zeolites
ENCS	:	On the inventory, or in compliance with the inventory
ISHL	:	On the inventory, or in compliance with the inventory
KECI	:	On the inventory, or in compliance with the inventory
PICCS	:	On the inventory, or in compliance with the inventory
IECSC	:	On the inventory, or in compliance with the inventory
NZIoC	:	Not in compliance with the inventory
EINECS	:	On the inventory, or in compliance with the inventory

15.2 Chemical safety assessment

No Chemical Safety Assessment has been carried out.

SECTION 16: Other information

Full text of other abbreviations

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - European Agreement concerning the International Carriage of Dangerous

Goods by Road; AICS - Australian Inventory of Chemical Substances; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of very high concern; TCSI - Taiwan Chemical Substance Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative

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